

SECTION C

10 (**short**) Three metal plates A, B, and C are placed in parallel with air between them, and all have an area of 200 cm^2 , as shown in Fig. 5. The distance between plates A and B is 2 mm, and the distance between plates B and C is 4 mm. Both plates A and C are connected to earth, but plate B has a total positive charge of $Q = 3.0 \times 10^{-7} \text{ C}$.

- (a) What are the induced charges in plates A and C? [5]
- (b) What is the potential V of plate B? [5]

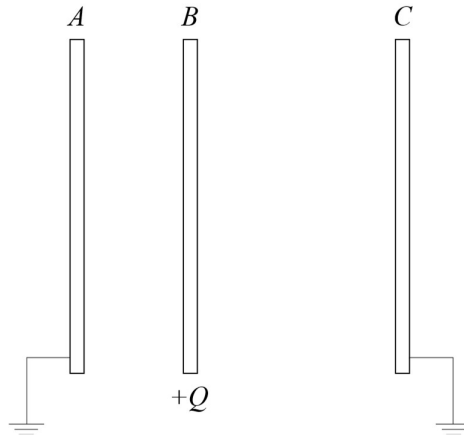


Fig. 5